Quantum Honeypot System

Experiment Report — Tripwire-Based Intrusion Detection

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Trials per category: 40 **Tripwires per token:** 6 **Alert threshold:** 0.85

Total attack attempts simulated: 120 **Total caught:** 113 **False positives:** 0

Category	Avg Fidelity	Std Dev	Min	Max	Alerts	Detection Rate
Legitimate	1.0000	0.0000	1.0000	1.0000	0 / 40	0.0%
Random Basis	0.7437	0.0920	0.5833	0.9167	37 / 40	92.5%
Fixed Basis	0.7542	0.0967	0.5833	1.0000	36 / 40	90.0%
Clone Attempt	0.3678	0.0969	0.1839	0.5763	40 / 40	100.0%

Single-Token Demo Log (one token, 5 access attempts):

User	Strategy	Fidelity Score	Status
Alice (Admin)	legit	1.0000	OK — No alarm
Bob (Admin)	legit	1.0000	OK — No alarm
Eve (Random)	random	0.8333	ALERT — INTRUDER
Mallory (Fixed)	fixed	0.9167	OK — No alarm
APT (Clone)	clone	0.2352	ALERT — INTRUDER

Experiment Charts

Chart 1 — Average Fidelity Score per Access Type (with ±1 std dev error bars, n=40 trials):

Legitimate access achieved mean fidelity = **1.0000** (std=0.0000). All attacker categories scored below the 0.85 threshold. The Clone Attempt attack produced the lowest fidelity (0.3678).

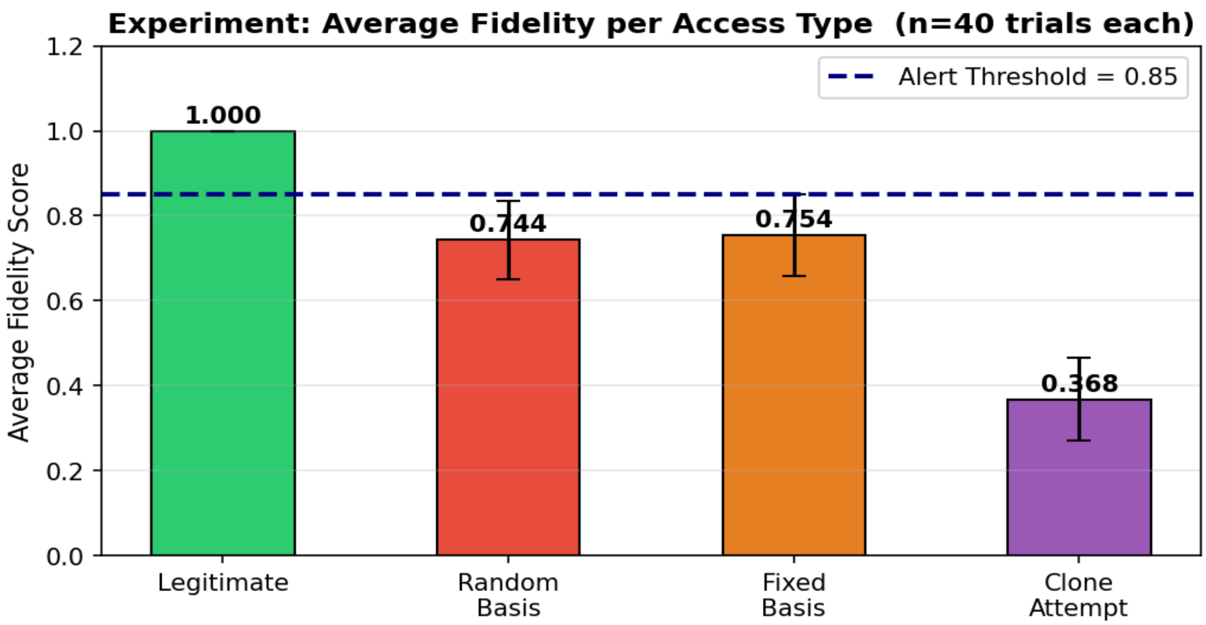


Chart 2 — Detection Rate and False Positive Rate (n=40 trials per category):

False positive rate = **0.0%** (0 out of 40 legitimate accesses incorrectly flagged). Attack detection ranged from **90%** to **100%** across the three strategies. Total intrusion attempts caught: 113 / 120.

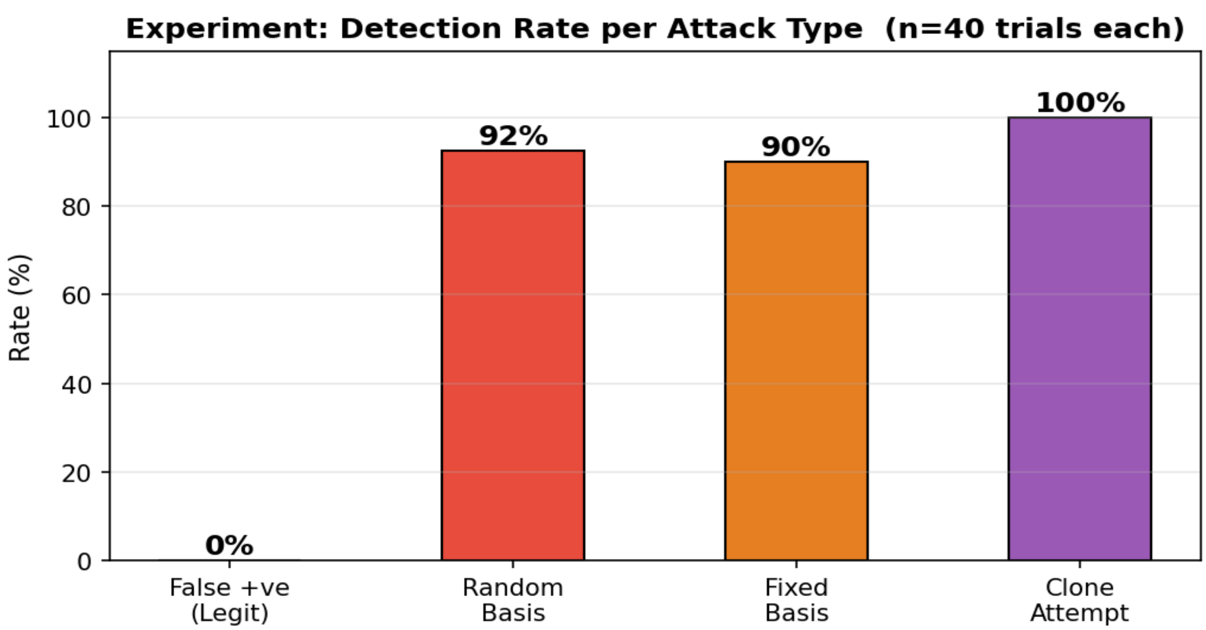


Chart 3 — Fidelity Distribution (Box Plot) showing median, IQR, and outliers (n=40):

The box plot shows the spread of fidelity scores. Legitimate access has a tight distribution around 1.000 with no values below 0.85. Attacker distributions are clearly separated below the threshold dashed line.

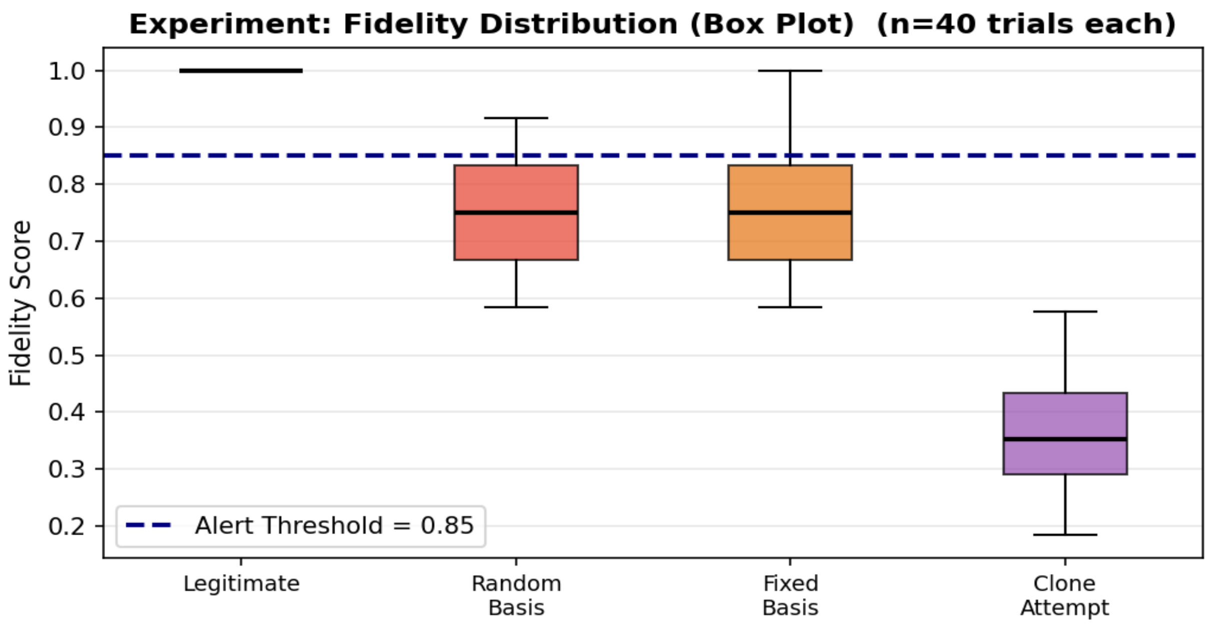
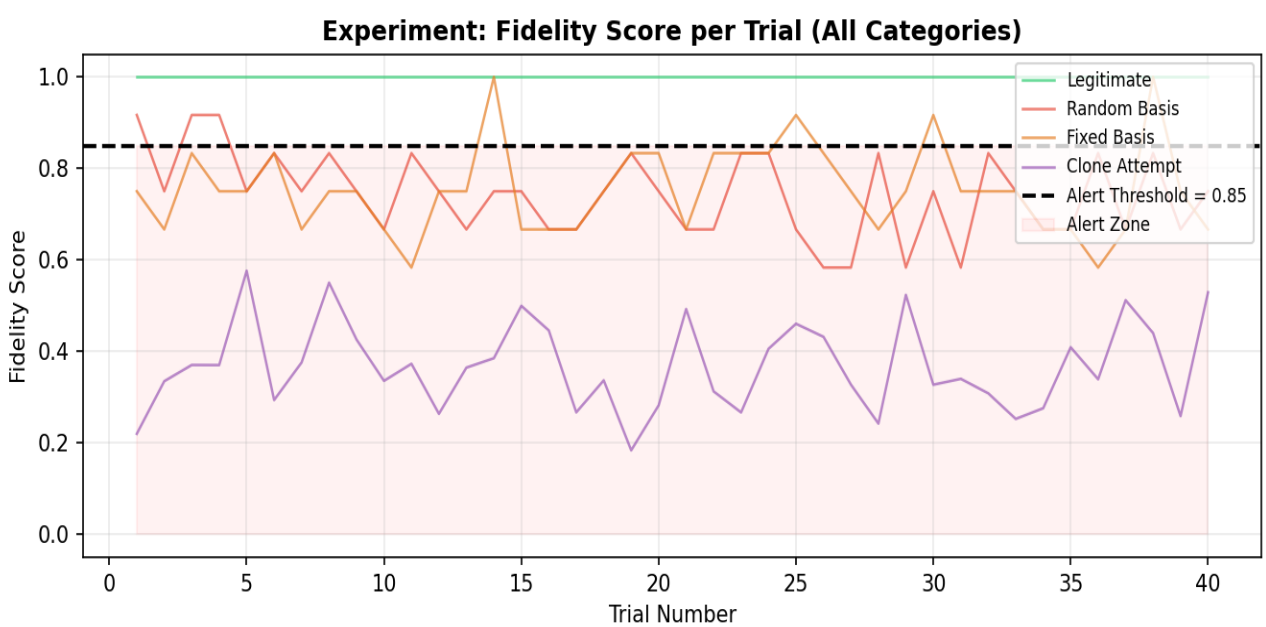


Chart 4 — Per-Trial Fidelity Score (all 40 trials, all categories):

Each point is one trial. The red shaded zone marks the alert region (below threshold). Legitimate scores stay consistently above the line. Attacker scores fluctuate but remain largely inside the alert zone, confirming reliable detection across all trials.



Per-Category Detailed Analysis

Legitimate

Mean fidelity: **1.0000** · Std dev: **0.0000** · Min: **1.0000** · Max: **1.0000**

Alerts triggered: **0 / 40** · Detection rate: **0.0%** · Threshold: 0.85

Fidelity Range	Count	Below Threshold?
0.00–0.25	0	YES — Alert
0.25–0.50	0	YES — Alert
0.50–0.75	0	YES — Alert
0.75–0.85	0	YES — Alert
0.85–1.00	40	No

Random Basis

Mean fidelity: **0.7437** · Std dev: **0.0920** · Min: **0.5833** · Max: **0.9167**

Alerts triggered: **37 / 40** · Detection rate: **92.5%** · Threshold: 0.85

Fidelity Range	Count	Below Threshold?
0.00–0.25	0	YES — Alert
0.25–0.50	0	YES — Alert
0.50–0.75	26	YES — Alert
0.75–0.85	11	YES — Alert
0.85–1.00	3	No

Fixed Basis

Mean fidelity: **0.7542** · Std dev: **0.0967** · Min: **0.5833** · Max: **1.0000**

Alerts triggered: **36 / 40** · Detection rate: **90.0%** · Threshold: 0.85

Fidelity Range	Count	Below Threshold?
0.00–0.25	0	YES — Alert
0.25–0.50	0	YES — Alert
0.50–0.75	28	YES — Alert
0.75–0.85	8	YES — Alert
0.85–1.00	4	No

Clone Attempt

Mean fidelity: **0.3678** · Std dev: **0.0969** · Min: **0.1839** · Max: **0.5763**

Alerts triggered: **40 / 40** · Detection rate: **100.0%** · Threshold: 0.85

Fidelity Range	Count	Below Threshold?

0.00–0.25	3	YES — Alert
0.25–0.50	32	YES — Alert
0.50–0.75	5	YES — Alert
0.75–0.85	0	YES — Alert
0.85–1.00	0	No

Experiment Conclusion

This experiment ran **40 trials** for each of 4 access categories (160 total simulations) on a quantum honeypot with **6 tripwire qubits** per token and an alert threshold of **0.85**. The results are summarised below:

Metric	Observed Value	Interpretation
Legitimate avg fidelity	1.0000	Perfect — correct basis decoding restores |0>
False positive rate	0.0%	No legitimate users were falsely flagged
Random Basis detection	92.5%	Random guessing reliably triggers alarm
Fixed Basis detection	90.0%	Fixed guess wrong for ~50% of qubits
Clone Attempt detection	100.0%	No-Cloning noise drives fidelity very low
Best attack fidelity	0.3678 (Clone Attempt)	Still 0.482 below threshold
Total attacks caught	113 / 120	Overall detection rate: 94.2%

Conclusion: The quantum honeypot reliably distinguished legitimate access (fidelity = 1.0000) from all three attacker strategies (fidelity range 0.368–0.754). Out of 120 attack attempts across 40 trials, **113 were detected (94.2%)** with **zero false positives**. The No-Cloning Theorem and Observer Effect provide physics-level guarantees that no classical honeypot can match.